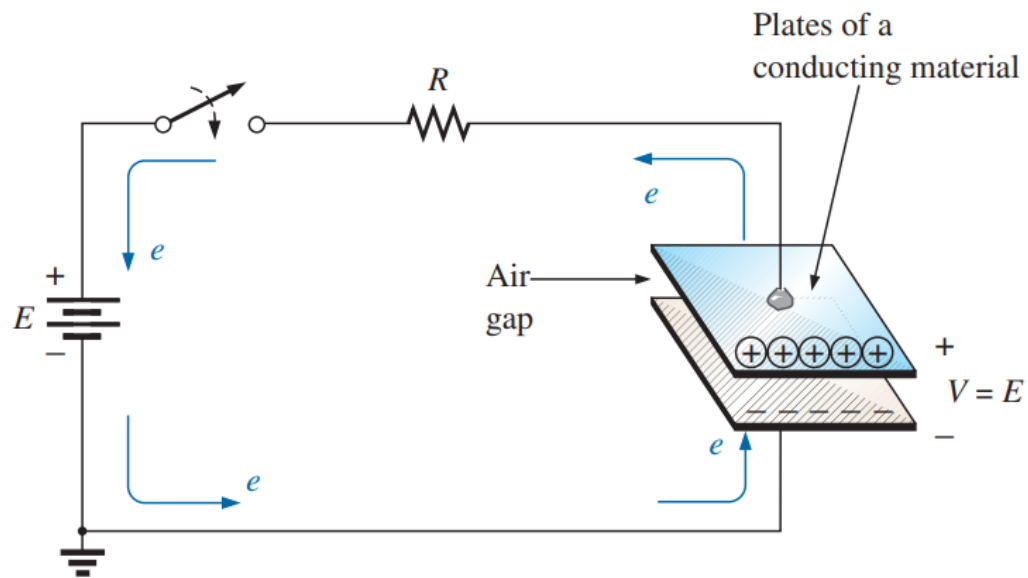


## Lecture 13

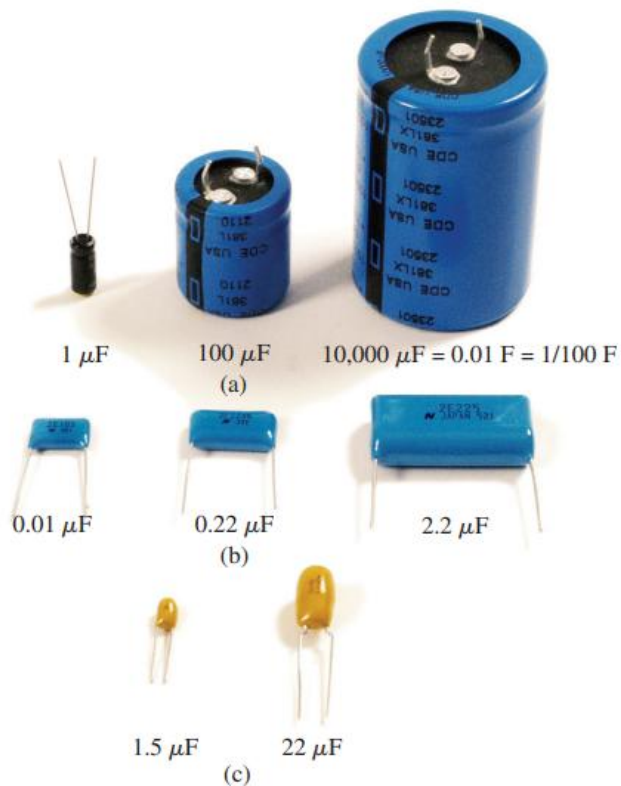
inductors and capacitors are energy-storage elements. They can store energy and later return it to the circuit. Capacitors and inductors do not generate energy—only the energy that has been put into these elements can be extracted.

Capacitors are constructed by separating two sheets of conductor, which is usually Capacitors are constructed by separating two conducting plates, which are usually metallic, by a thin layer of insulating material. metallic, by a thin layer of insulating material. In a parallel-plate capacitor, the sheets are flat and parallel as shown in Figure 3.1. The insulating material between the plates, called a dielectric, can be air, Mylar®, polyester, polypropylene, mica, or a variety of other materials.



**FIG. 10.4**  
*Fundamental charging circuit.*

- Capacitance is a measure of a capacitor's ability to store charge on its plates—in other words, its storage capacity.
- the higher the capacitance of a capacitor, the greater is the amount of charge stored on the plates for the same applied voltage.
- The unit of measure applied to capacitors is the farad (F), named after an English scientist, Michael Faraday, who did extensive research in the field (Fig. 10.5). In particular, a capacitor has a capacitance of 1 F if 1 C of charge ( $6.242 \times 10^{18}$  electrons) is deposited on the plates by a potential difference of 1 V across its plates



**FIG. 10.12**

*Demonstrating that, in general, for each type of construction, the size of a capacitor increases with the capacitance value: (a) electrolytic; (b) polyester-film; (c) tantalum.*

$$q = Cv$$

$$q(t) = \int_{t_0}^t i(t) dt + q(t_0)$$

$$i = \frac{dq}{dt} = \frac{d}{dt}(Cv)$$

$$v(t) = \frac{1}{C} \int_{t_0}^t i(t) dt + \frac{q(t_0)}{C}$$

$$i = C \frac{dv}{dt}$$

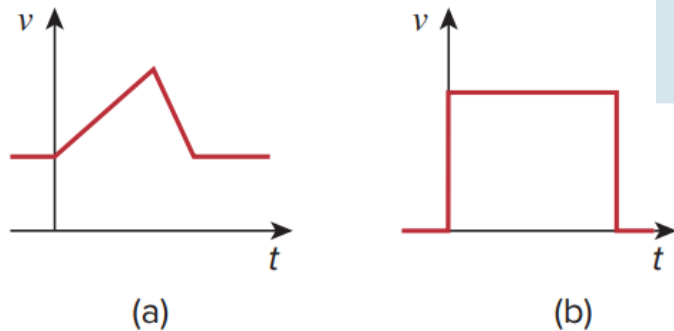
$$v(t) = \frac{1}{C} \int_{t_0}^t i(t) dt + v(t_0)$$

$$p = vi = Cv \frac{dv}{dt}$$

$$\int_{t_0}^t p \, dt' = C \int_{t_0}^t v \frac{dv}{dt'} \, dt' = C \int_{v(t_0)}^{v(t)} v' \, dv' = \frac{1}{2} C \{[v(t)]^2 - [v(t_0)]^2\}$$

$$w_C(t) - w_C(t_0) = \frac{1}{2} C \{[v(t)]^2 - [v(t_0)]^2\}$$

$$i = C \frac{dv}{dt}$$

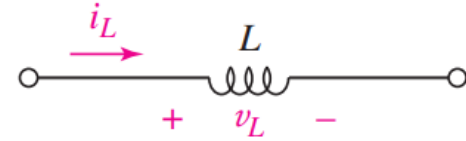
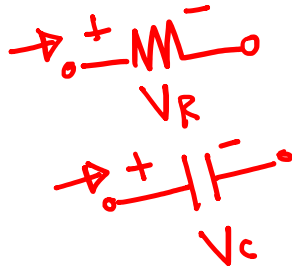


A capacitor is an open circuit to dc.

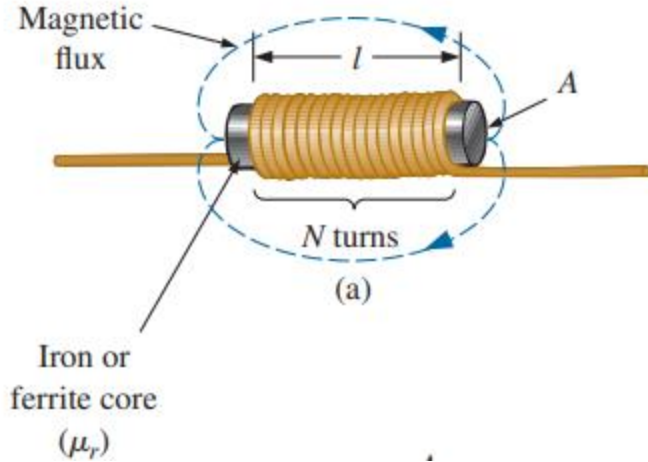
### Figure 6.7

Voltage across a capacitor: (a) allowed, (b) not allowable; an abrupt change is not possible.

The voltage on a capacitor cannot change abruptly.

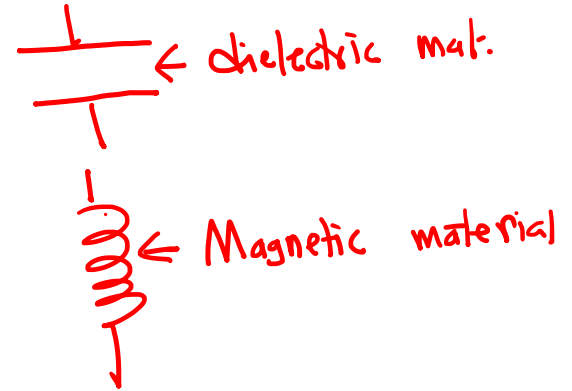


**FIGURE 7.10** Electrical symbol and current-voltage conventions for an inductor.



Magnetic materi.

air.



- A current carrying conductor produces a magnetic field. (Oerstad)
- Magnetic field was linearly related to the current which produced it. (Ampere)
- A changing magnetic field could induce a voltage in a neighboring circuit whose polarity is such that it opposes the causes that (Faraday/Henry)
- This voltage was proportional to the time rate of change of the current producing the magnetic field. (Faraday/Henry)

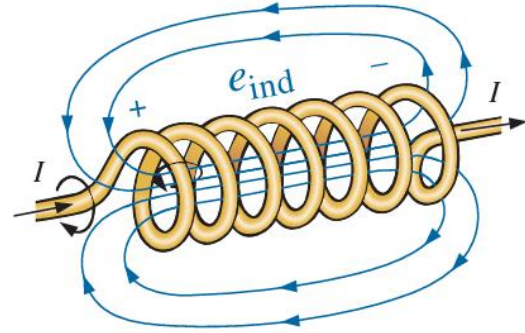
**The constant of proportionality is what we now call the inductance, symbolized by L.**

$$v = L \frac{di}{dt}$$

Inductance is the property whereby an inductor exhibits opposition to the change of current flowing through it, measured in henrys (H).

$$i(t) = \frac{1}{L} \int_{t_0}^t v dt' + i(t_0)$$

The inductance of a coil is also a measure of the change in flux linking the coil due to a change in current through the coil.



$$L = N \frac{d\phi}{di_L}$$

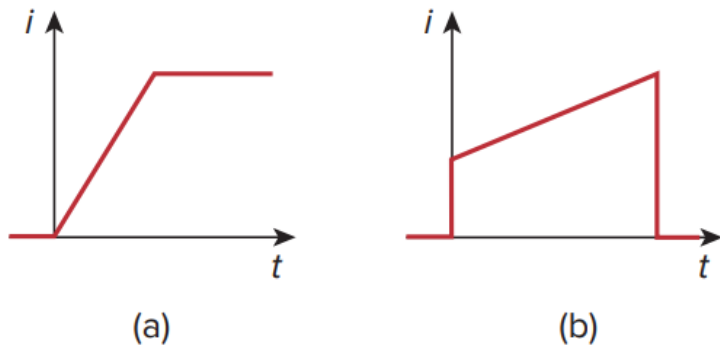
$$e = N \frac{d\phi}{dt} = \left( N \frac{d\phi}{di_L} \right) \left( \frac{di_L}{dt} \right)$$

$$e_L = L \frac{di_L}{dt}$$

$$v = L \frac{di}{dt}$$

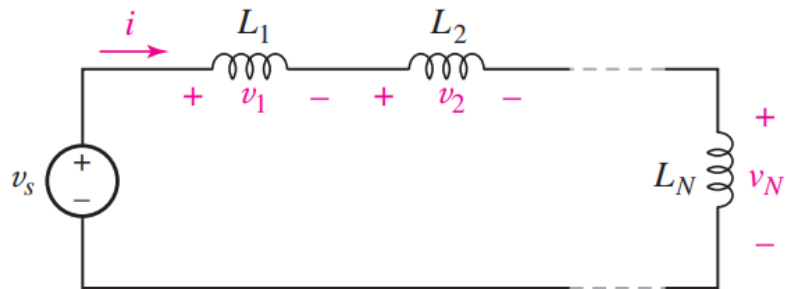
An inductor acts like a short circuit to dc.

The current through an inductor cannot change instantaneously.

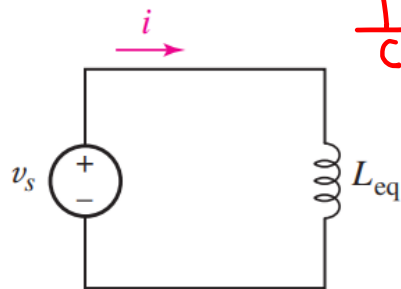


**Figure 6.25**

Current through an inductor: (a) allowed, (b) not allowable; an abrupt change is not possible.



(a)



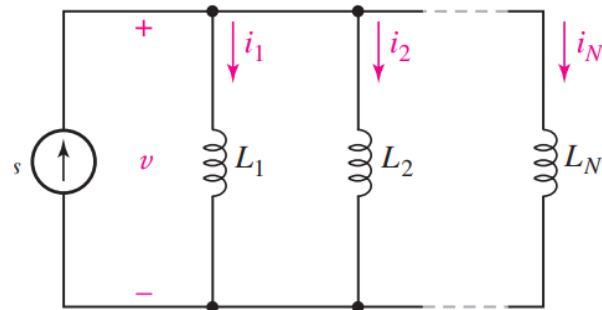
(b)

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_N}$$

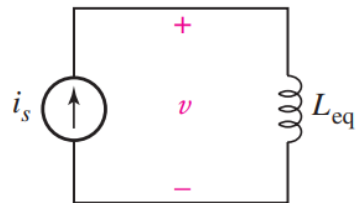
$$L_{eq} = L_1 + L_2 + L_3 + \dots + L_N$$

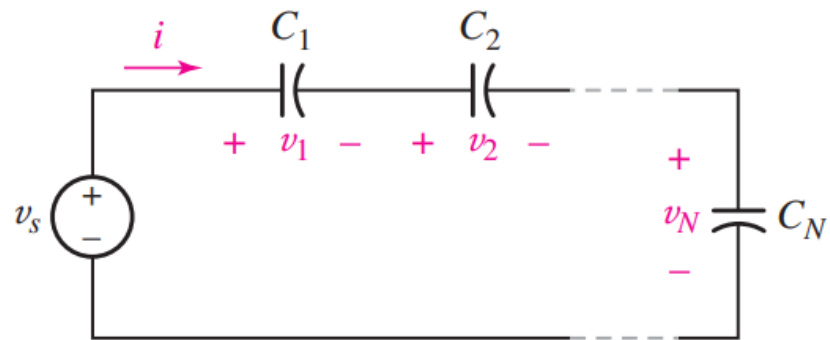
$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \dots + \frac{1}{L_N}$$

$$C_{eq} = C_1 + C_2 + C_3$$

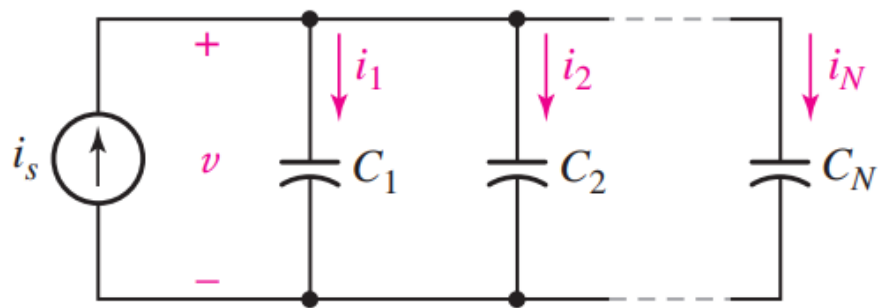


(a)

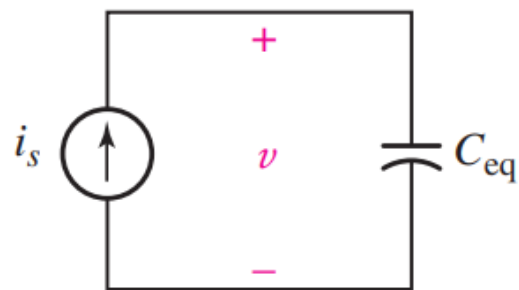
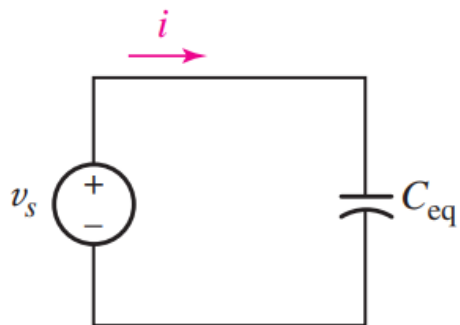




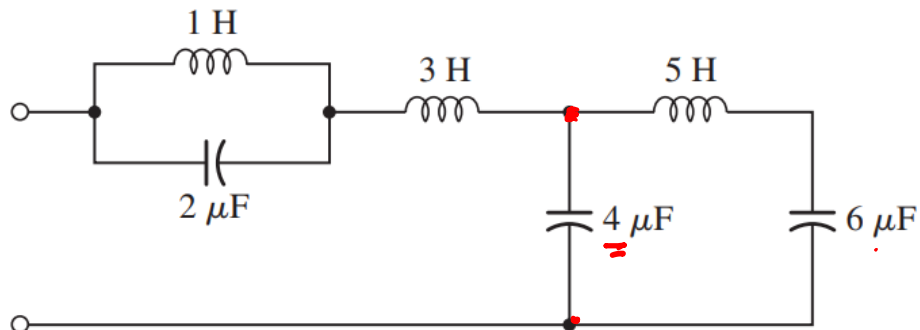
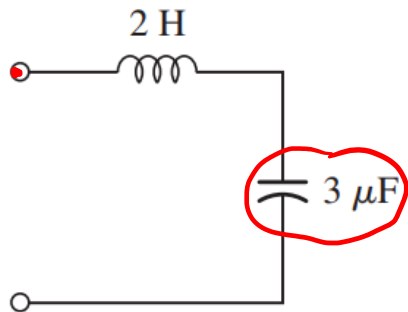
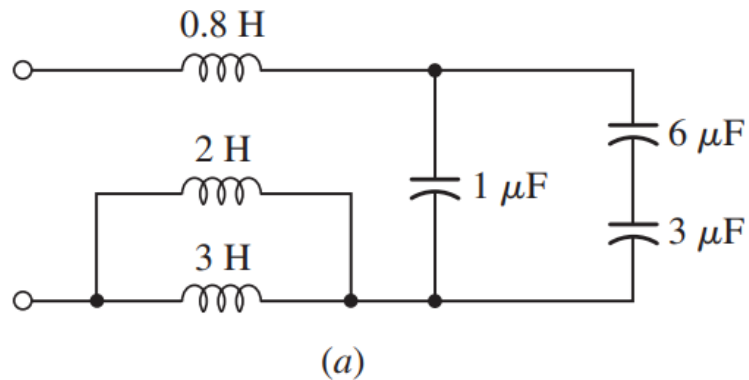
(a)



(a)



(b)



■ **FIGURE 7.24** An LC network in which no series or parallel combinations of either the inductors or the capacitors can be made.

$$\frac{1}{6} + \frac{1}{3} = \frac{18}{9} = 2 \mu F$$

$$\frac{6}{5} = 1.2 + 0.8$$

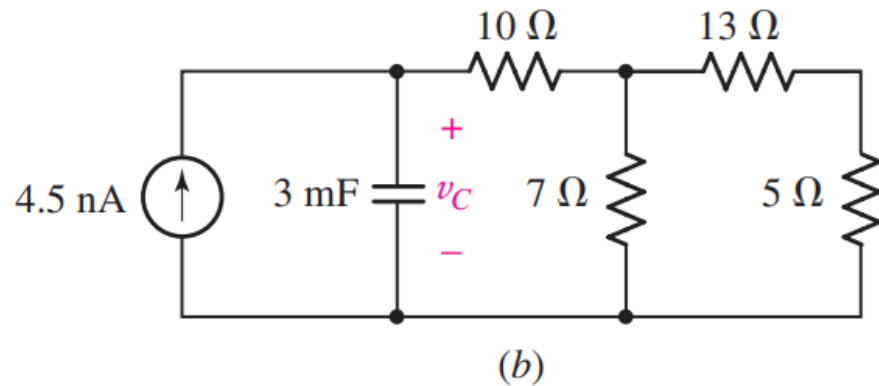
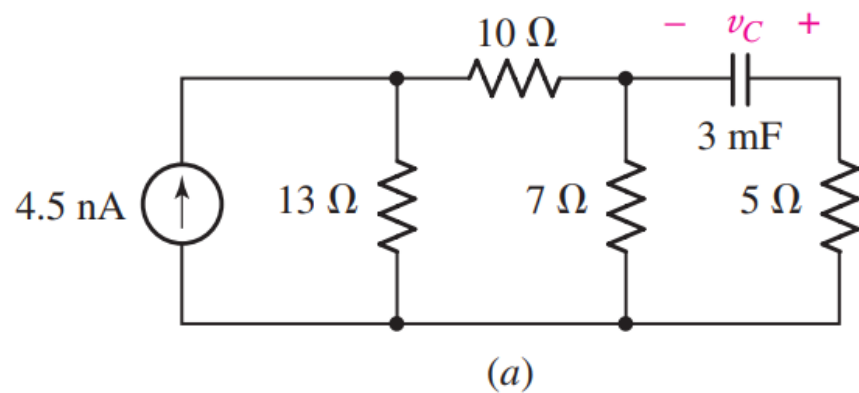
$$2 + 1 = 3 \mu H$$

TABLE 6.1

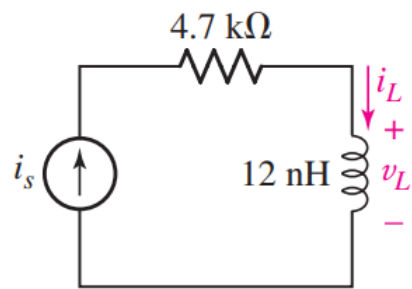
Important characteristics of the basic elements.<sup>†</sup>

| Relation                                            | Resistor ( $R$ )                     | Capacitor ( $C$ )                                                 | Inductor ( $L$ )                                                              |
|-----------------------------------------------------|--------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <u><math>v</math>-<math>i</math>:</u>               | $v = iR$                             | $v = \frac{1}{C} \int_{t_0}^t i(\tau) d\tau + \underline{v(t_0)}$ | $v = L \frac{di}{dt}$                                                         |
| $i$ - $v$ :                                         | $i = v/R$                            | $i = C \frac{dv}{dt}$                                             | $i = \frac{1}{L} \int_{t_0}^t v(\tau) d\tau + \underline{i(t_0)} \rightarrow$ |
| $p$ or $w$ :                                        | $p = i^2 R = \frac{v^2}{R}$          | $w = \frac{1}{2} C v^2$                                           | $w = \frac{1}{2} L i^2$                                                       |
| Series:                                             | $R_{eq} = R_1 + R_2$                 | $C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$                              | $L_{eq} = L_1 + L_2$                                                          |
| Parallel:                                           | $R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$ | $C_{eq} = C_1 + C_2$                                              | $L_{eq} = \frac{L_1 L_2}{L_1 + L_2}$                                          |
| At dc:                                              | Same $\neq$                          | Open circuit $\neq$                                               | Short circuit                                                                 |
| Circuit variable<br>that cannot<br>change abruptly: | Not applicable $\underline{v}$       |                                                                   | $\underline{i}$                                                               |

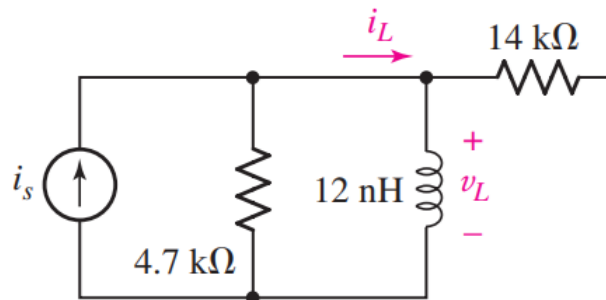
15. For each circuit shown in Fig. 7.45, calculate the voltage labeled  $v_C$ .



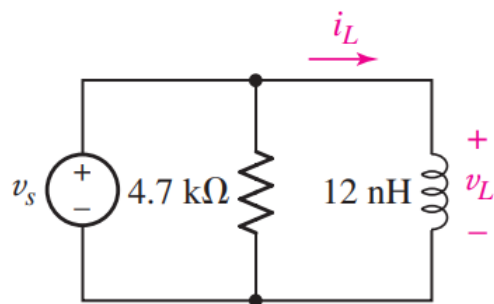
21. Calculate  $v_L$  and  $i_L$  for each of the circuits depicted in Fig. 7.48, if  $i_s = 1$  mA and  $v_s = 2.1$  V.



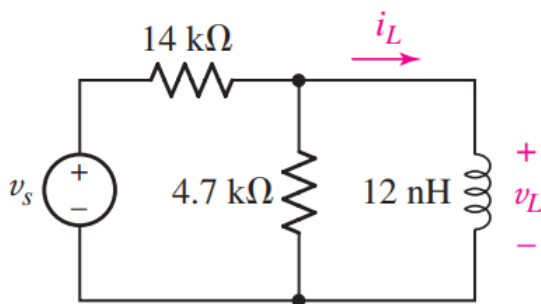
(a)



(b)



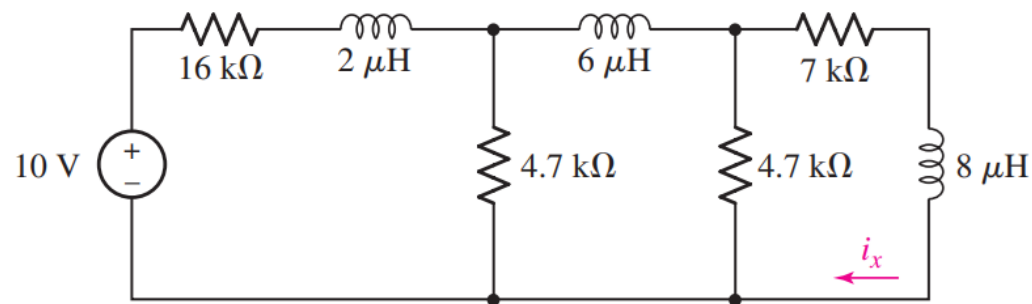
(c)



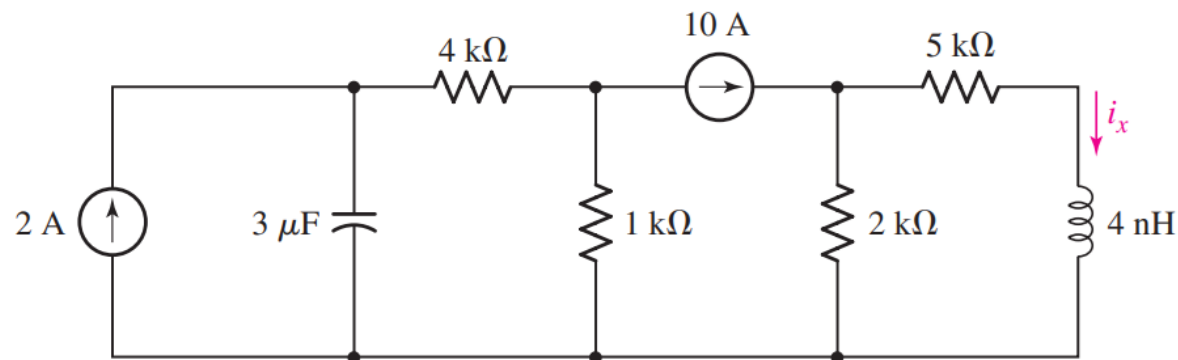
(d)

■ FIGURE 7.48

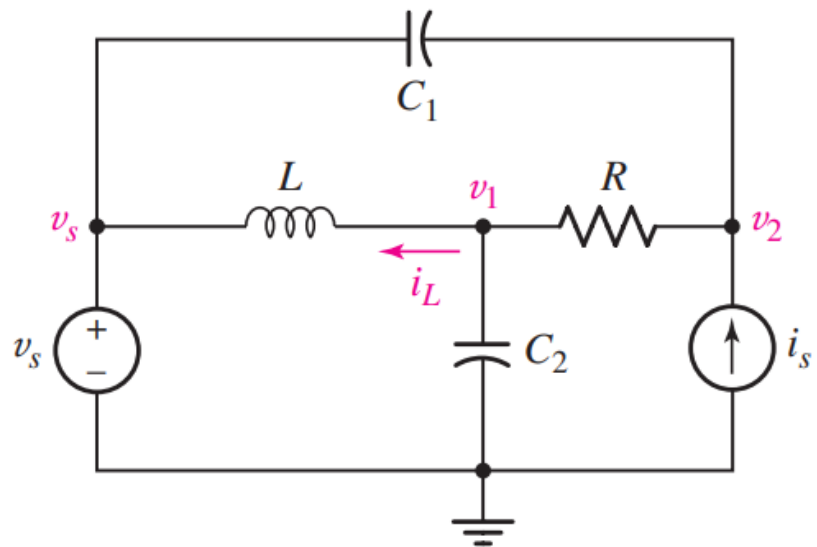
28. Making the assumption that the circuits in Fig. 7.50 have been connected for a very long time, determine the value for each current labeled  $i_x$ .

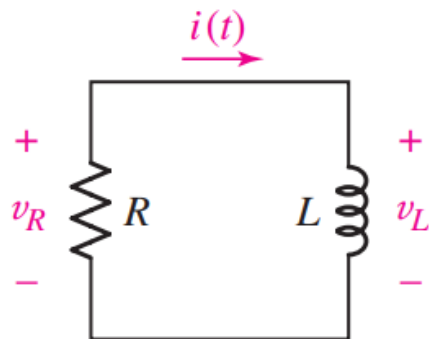


(a)



(b)





■ **FIGURE 8.1** A series  $RL$  circuit for which  $i(t)$  is to be determined, subject to the initial condition that  $i(0) = I_0$ .

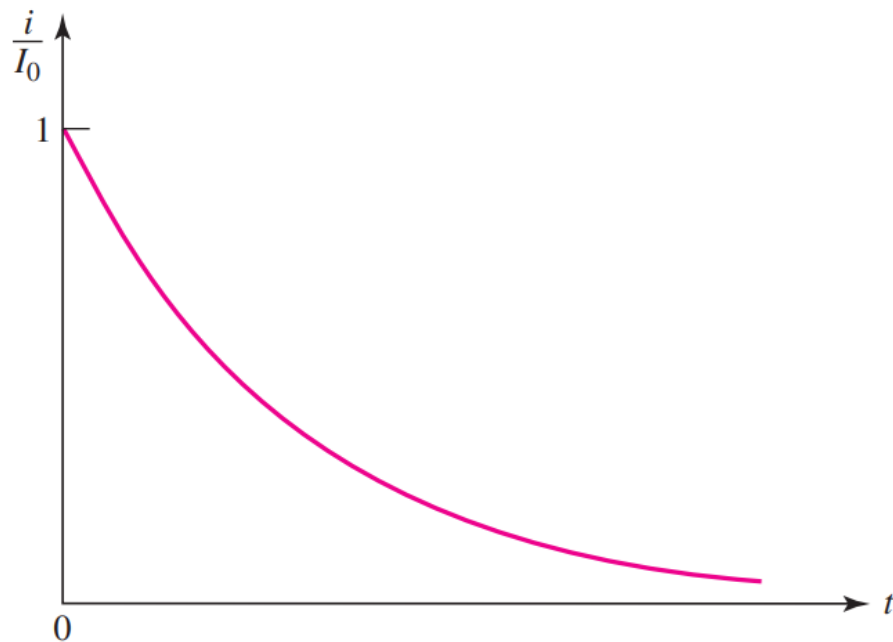
$$v_R + v_L = 0$$

$$Ri + L \frac{di}{dt} = 0$$

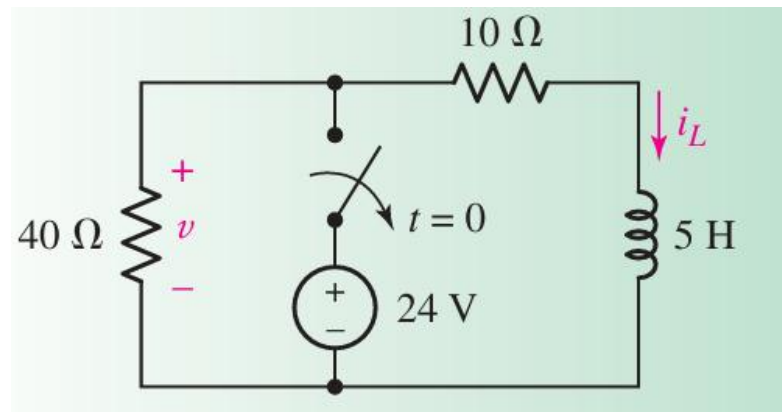
$$i + \frac{L}{R} \frac{di}{dt} = 0$$

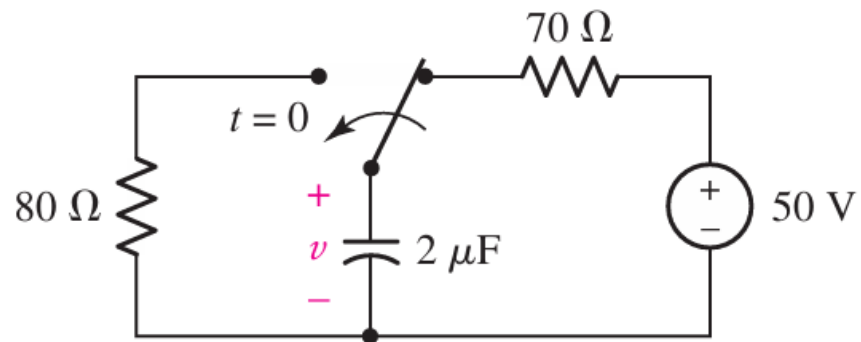
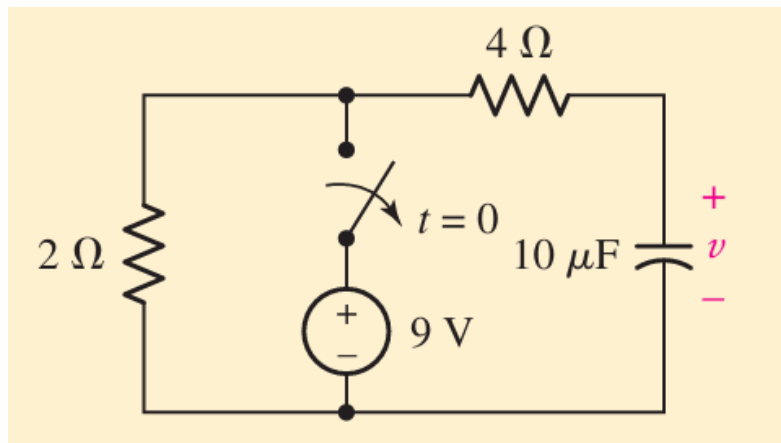
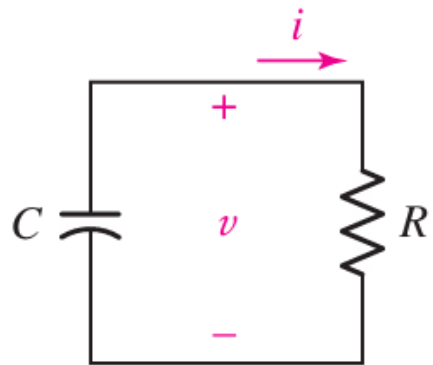
$$i = I_0 e^{-\frac{t}{L/R}}$$

$$i = I_0 e^{-\frac{t}{\tau}}$$



■ **FIGURE 8.7** A plot of  $e^{-Rt/L}$  versus  $t$ .

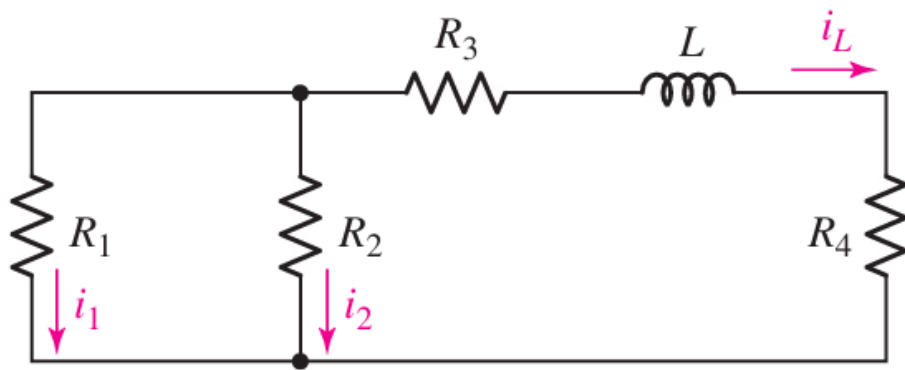


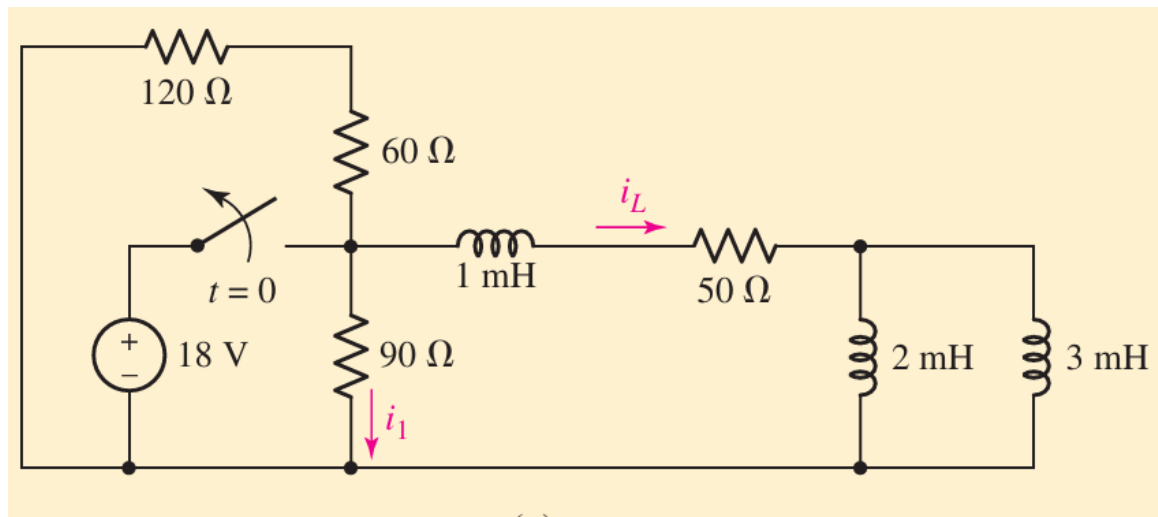


The **natural response** of a circuit refers to the behavior (in terms of voltages and currents) of the circuit itself, with no external sources of excitation.

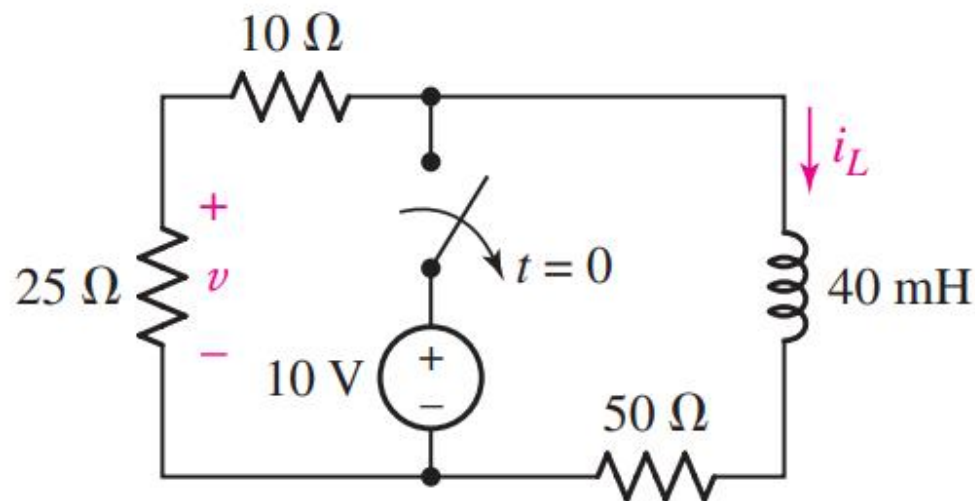
The **time constant** of a circuit is the time required for the response to decay to a factor of  $1/e$  or 36.8 percent of its initial value.<sup>1</sup>

## General *RL* Circuits



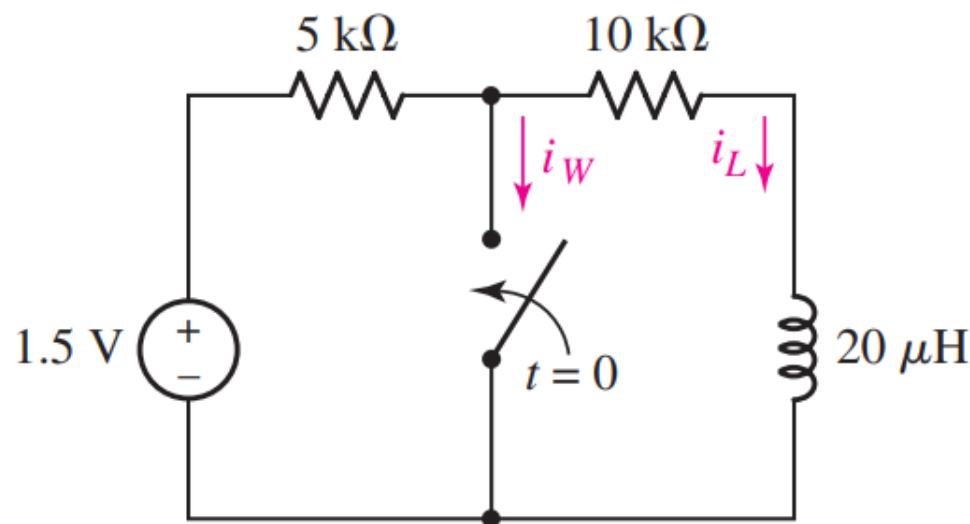


9. The switch in the circuit of Fig. 8.55 has been closed a ridiculously long time before suddenly being thrown open at  $t = 0$ . (a) Obtain expressions for  $i_L$  and  $v$  in the circuit of Fig. 8.55 which are valid for all  $t \geq 0$ . (b) Calculate  $i_L(t)$  and  $v(t)$  at the instant just prior to the switch opening, at the instant just after the switch opening, and at  $t = 470 \mu\text{s}$ .



■ FIGURE 8.55

10. Assuming the switch initially has been open for a really, really long time, (a) obtain an expression for  $i_W$  in the circuit of Fig. 8.56 which is valid for all  $t \geq 0$ ; (b) calculate  $i_W$  at  $t = 0$  and  $t = 1.3$  ns.

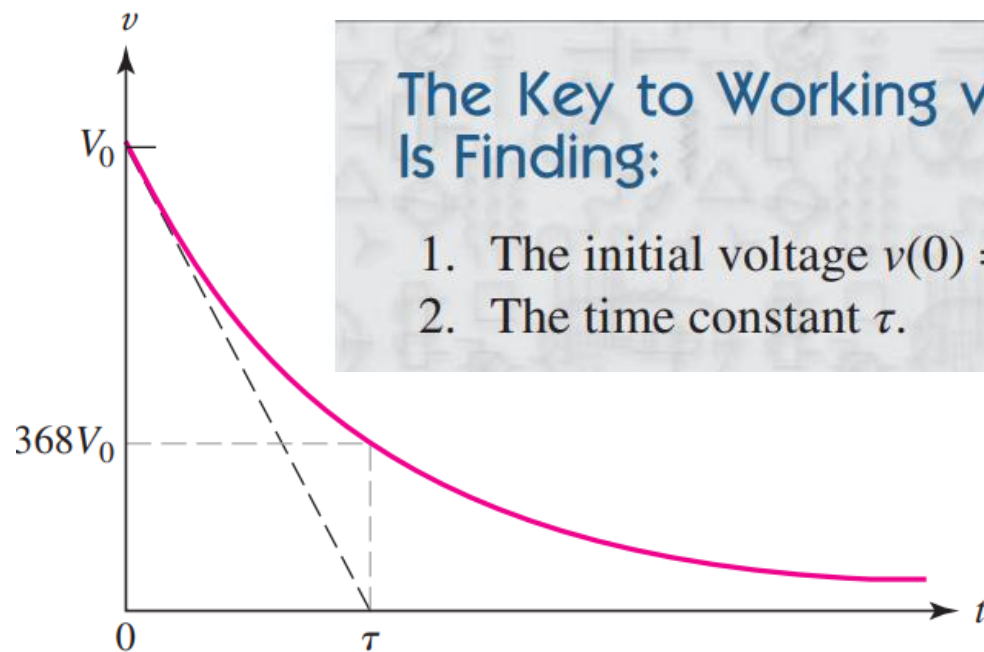


■ FIGURE 8.56

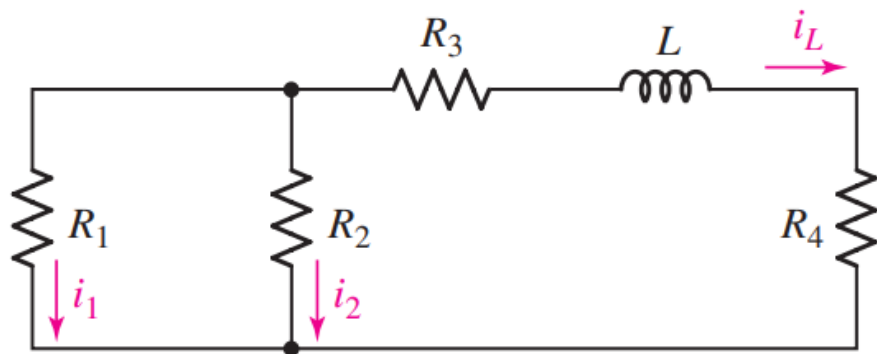
$$\tau = RC$$

The Key to Working with a Source-Free  $RC$  Circuit Is Finding:

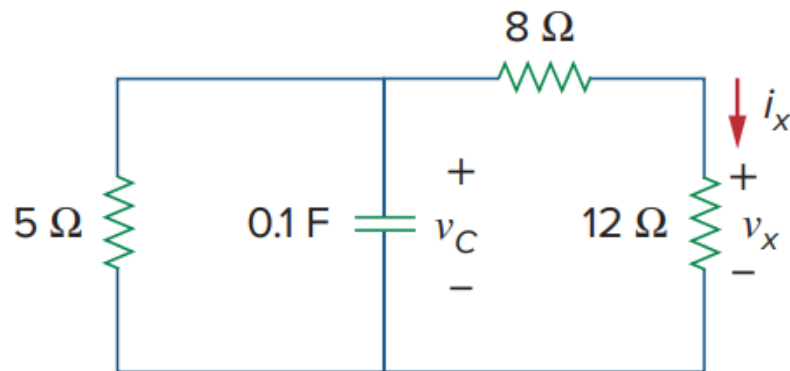
1. The initial voltage  $v(0) = V_0$  across the capacitor.
2. The time constant  $\tau$ .



**FIGURE 8.16** The capacitor voltage  $v(t)$  in the parallel  $RC$  circuit is plotted as a function of time. The initial value of  $v(t)$  is  $V_0$ .



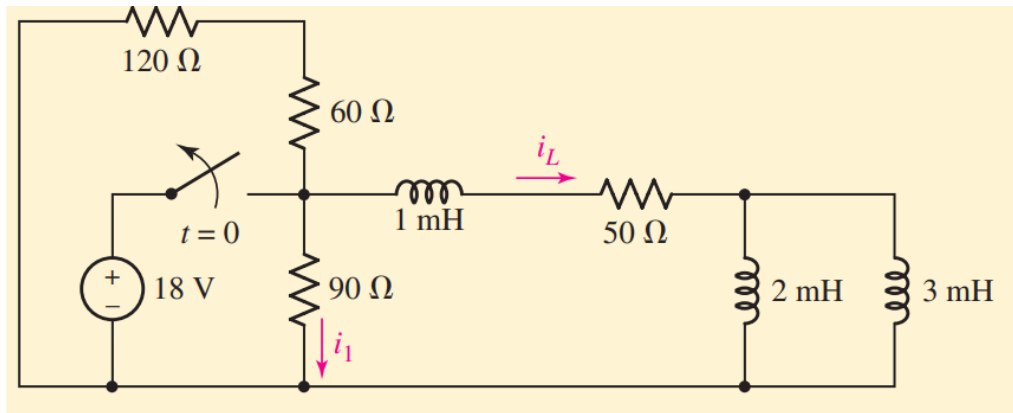
■ **FIGURE 8.19** A source-free circuit containing one inductor and several resistors is analyzed by determining the time constant  $\tau = L/R_{\text{eq}}$ .



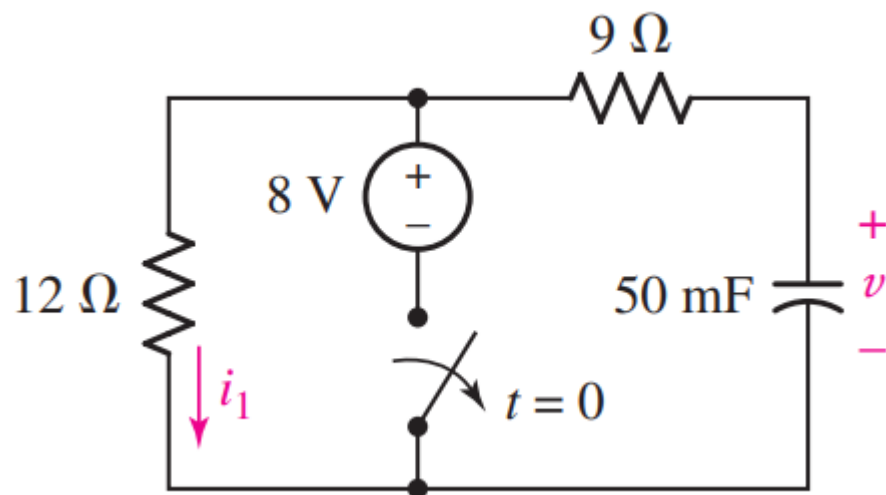
**Figure 7.5**  
For Example 7.1.

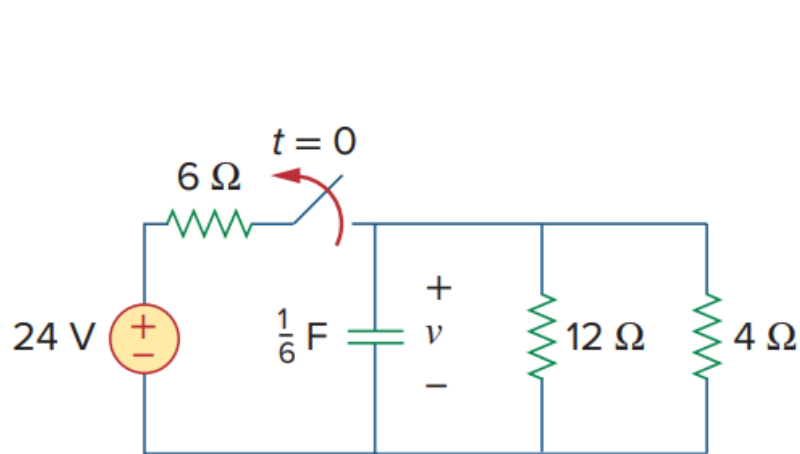
In Fig. 7.5, let  $v_C(0) = 15\text{ V}$ . Find  $v_C$ ,  $v_x$ , and  $i_x$  for  $t > 0$ .

find the equivalent resistance or the Thevenin Resistance at the capacitor terminals.



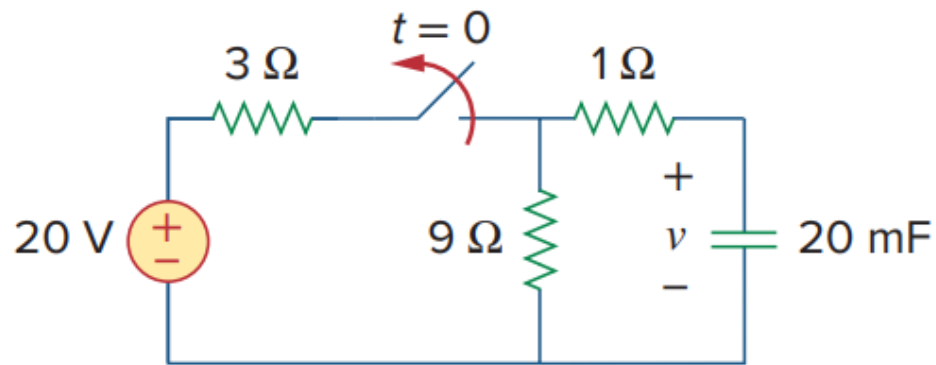
21. We can safely assume the switch in the circuit of Fig. 8.59 was closed a very long time prior to being thrown open at  $t = 0$ . (a) Determine the circuit time constant. (b) Obtain an expression for  $i_1(t)$  which is valid for  $t > 0$ . (c) Determine the power dissipated by the  $12\ \Omega$  resistor at  $t = 500\text{ ms}$ .



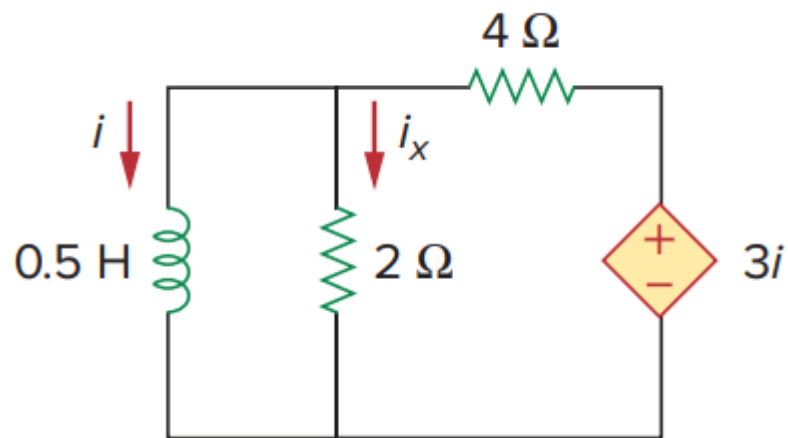


**Figure 7.10**

For Practice Prob. 7.2.

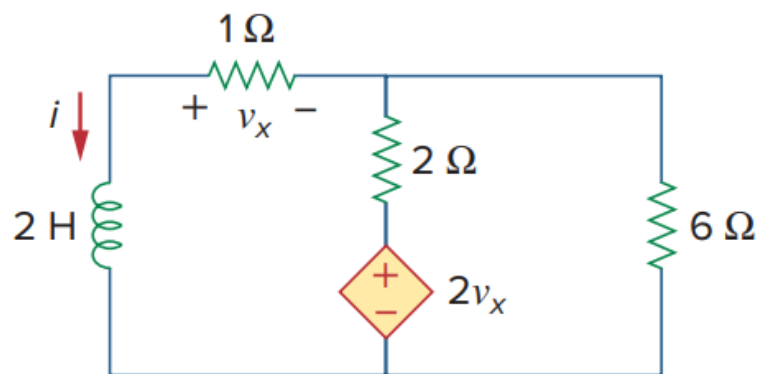


The switch in the circuit in Fig. 7.8 has been closed for a long time, and it is opened at  $t = 0$ . Find  $v(t)$  for  $t \geq 0$ . Calculate the initial energy stored in the capacitor.



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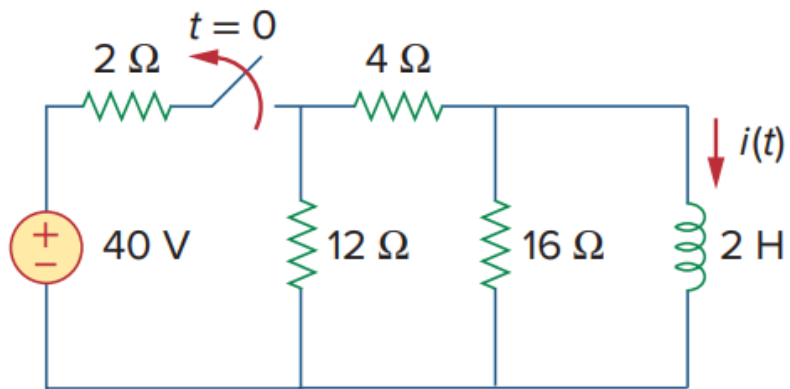
Assuming that  $i(0) = 10\text{ A}$ , calculate  $i(t)$  and  $i_x(t)$  in the circuit of Fig. 7.13.



**Figure 7.15**

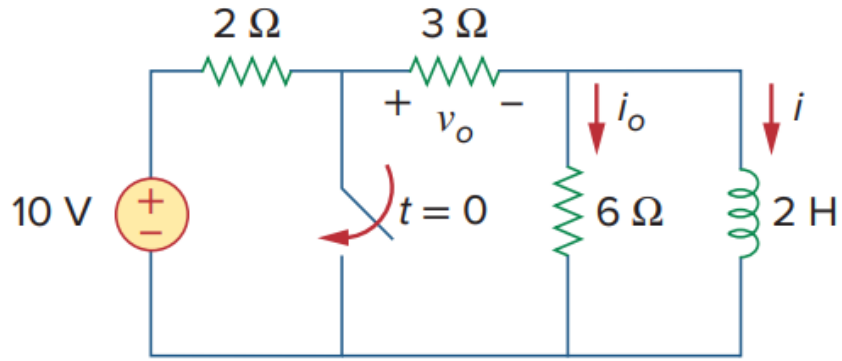
For Practice Prob. 7.3.

Find  $i$  and  $v_x$  in the circuit of Fig. 7.15. Let  $i(0) = 7$  A.



**Figure 7.16**

For Example 7.4.



In the circuit shown in Fig. 7.19, find  $i_o$ ,  $v_o$ , and  $i$  for all time, assuming that the switch was open for a long time

✓

The **natural response** of a circuit refers to the behavior (in terms of voltages and currents) of the circuit itself, with no external sources of excitation.

The **time constant** of a circuit is the time required for the response to decay to a factor of  $1/e$  or 36.8 percent of its initial value.<sup>1</sup>